Supplementary file 3 Table S1 Characteristics of included trials and patients.

Study (Year)	Inclusion criteria	Size of study	Men (%)	Mean age, years	Type of study	Follow-up, months	Outcome	Treatment group	Definition and detail of salt reduction	Measurement of salt intake	Control group	Mean eGFR	Mean blood pressure (mmHg)	Funding Source
Mühlhauser I[18] (1996)	Insulin-dependent diabetic patients with nephropathy and high normal or mildly hypertensive blood pressure	16	75	35	Randomised parallel-group controlled trial	1	UPE, SBP, DBP	Reduce sodium intake to about 90 mmol/d	Reduce sodium intake to about 90 mmol/day	Salt intake were estimated by analysis of the dietary protocol and analysis of urine	Sodium intake about 190 mmol/d	87.9[a]	132/101	Non- industry
Cianciaruso B[19] (1998)	Ccr 10–40 mL/min	57	72	50	Cohort study	42.95	Rate of change in eGFR, UPE	Urinary sodium < 100 mmol/d	Urinary sodium < 100 mmol/d	24-h urinary sodium	Urinary sodium > 200 mmol/d	38	NA	NA
Keven K[20] (2006)	Patients underwent kidney transplantation and had stable allograft function	32	78	41	Randomised parallel-group controlled trial	3	SBP, DBP	Sodium intake 80-100 mmol/d	Sodium intake 80-100 mmol/d	24-h urinary sodium	Usual care	1.39c	143/88	NA
Fujiwara S[21] (2010)	UACR ≥ 30 mg/g Cr	36	56	73	Cohort study	3	UACR, SBP, DBP	Intensive low-salt diet education	NA	salt intake was examined, using Excel Eiyokun (version 4.5, KENPAKUSHA, Tokyo, Japan) for the assessment of dietary intake by food frequency questionnaire.	Usual care	65.6	140/67	Non- industry
Yu W[22] (2011)	Immunoglobulin A nephropathy	43	53	36	Cohort study	0.25	UPE, SBP, DBP	Sodium intake 100mmol/d	Sodium intake 100mmol/d	24-h urinary sodium	Usual diet	70.2	118/75	NA
Vegter S[23] (2012)	CKD patients without diabetes	500	76	52	Cohort study	51	Renal failure	Urinary sodium/creatinine < 200 mmol/g	Urinary sodium/creatinine < 200 mmol/g	Urinary sodium/creatinine	Urinary sodium/creatinine≥ 200 mmol/g	42.5[b]	144/95	Non- industry

Heerspink HJL[24] (2012)	Type 2 diabetic nephropathy	1177	65	59	Cohort study	30	Renal composite outcomes, CV events	Urinary sodium/creatinine< 153mmol/g	Urinary sodium/creatinine< 153mmol/g	Urinary sodium/creatinine	Urinary sodium/creatinine≥ 153mmol/g	44	155/84	Industry and non- industry
LowSALT CKD Study[25,26] (2012)	Hypertensive patients with stage 3–4 CKD	20	75	69	Randomised crossover trial	1	UACR, SBP, DBP	Sodium intake < 100 mmol/d	Sodium intake < 100 mmol/d	24-h urinary sodium	Sodium intake ≥ 220 mmol/d	32	151/82	Non- industry
de Brito-Ashurst I[27] (2013)	eGFR < 60 mL/min and mean BP > 130/80 mm Hg	48	58	58	Randomised parallel-group controlled trial	6	Change in eGFR	Intensive low-salt diet education	Traditional meals had salt reduced by 50%	24-h urinary sodium	Usual care	41.5	153/85	Non- industry
Liang X[28] (2013)	PD	106	54	33	Cohort study	1	CV events, SBP, DBP	Health education to restrict water-sodium intake strictly	Salt intake not more than 3 g/d	The 3-day diet diary was collected once a month by the special nurse. Compared with the Food Composition, the daily salt intake were estimated	Usual care	NA	149/88	NA
Mc Quarrie EP[29] (2014)	Patients with CKD and had at least one 24-hour urinary sodium measurement	423	50	52	Cohort study	102	Renal composite outcomes, all- cause mortality	Urinary sodium/creatinine ≤ 16	Urinary sodium/creatinine ≤ 16	Urinary sodium/creatinine	Urinary sodium/creatinine > 16	48	139/79	Non- industry
Fan L[4] (2014)	GFR 13–55 mL/min/1.73m^2	840	60	52	Cohort study	72	Renal composite outcomes, all- cause mortality	Urinary sodium < 3 g/d	Urinary sodium < 3 g/d	24-h urinary sodium	Urinary sodium ≥ 3 g/d	32.5	132/81	Non- industry
Baek SH[30] (2014)	Hypertensive patients with albuminuria	202	51	50	Cohort study	2	UAE, SBP, DBP	Good compliance with an low salt diet	Good compliance with an low salt diet (24-hurinary sodium reduction ratio < 50%)	24-hurinary sodium	Poor compliance with an low salt diet	77.1[c]	121/73	Industry and non- industry

Hwang JH[31] (2014)	eGFR ≥ 30 mL/min/1.73m2 and UACR ≥ 30 mg/g	245	50	50	Randomised parallel-group controlled trial	2	eGFR, UAE, SBP, DBP	Sodium intake < 100mmol/d	Sodium intake < 100mmol/d	24-hurinary sodium	Usual care	67.3	131/79	Industry
Rodrigues Telini LS[32] (2014)	HD	39	38	58	Randomised parallel-group controlled trial	4	SBP, DBP	2 g of sodium reduction in their habitual diet	2 g of sodium reduction in their habitual diet	Salt intake was based on a 72-halimentary registry	Usual diet	NA	146/86	Non- industry
Nerbass FB[33] (2015)	eGFR 59–30 ml/min/1·73 m^2	1416	39	73	Cohort study	12	eGFR, UACR, SBP, DBP	Sodium intake decrease	Sodium intake decrease	Estimated 24-hurinary sodium	Sodium intake unchanged or increase	52.7	134/73	Industry and non- industry
CRIC Study[3,34] (2016)	eGFR20-70mL/min/1.73 m^2	3757	56	58	Cohort study	78	Renal composite outcomes, all- cause mortality, CV events	Urinary sodium < 153.6mmol/d	Urinary sodium < 153.6mmol/d	24-hurinary sodium	Urinary sodium ≥153.7 mmol/d	44.5	128/71	Non- industry
Mazarova A[5] (2016)	eGFR < 30 mL/min/1.73 m^2	341	62	64	Cohort study	18	Change in eGFR, renal failure, all-cause mortality	Urinary sodium ≤ 150mmol/d	Urinary sodium ≤ 150mmol/d	24-hurinary sodium	Urinary sodium> 150 mmol/d	17.1	NA	No funding supported
Kauric-Klein Z[35] (2017)	HD	121	50	60	Randomised parallel-group controlled trial	3	All- cause mortality, SBP, DBP	Sodium intake < 2g/d	Sodium intake < 2g/d	Sodium intake checklist developed by Millar and Beard	Usual care	NA	165/87	Non- industry
Meuleman Y[36] (2017)	eGFR ≥ 20mL/min/1.73m^2 and UPE > 0.2 g/L	112	82	55	Randomised parallel-group controlled trial	6	eGFR, UPE, SBP, DBP	Usual care and low salt diet education	low salt diet education	24-hurinary sodium	Usual care	49.7	129/76	Non- industry

Jiamjariyapon T[37] (2017)	Stage 3-4 CKD with diabetes or hypertension	418	27	62	Cluster randomized controlled trial	24	Renal composite outcomes, all-cause mortality, eGFR, CV events, UACR, SBP, DBP	Usual care and intensive low-salt diet education	Intensive low-salt diet educatio (Sodium intake less than 2,400 mg/d)	24-hurinary sodium	Usual care	41.5	125/76	Non- industry
Saran R[38] (2017)	eGFR 15–60 ml/min/1.73 m^2	58	60	61	Randomized crossover trial	1	eGFR, UACR, SBP, DBP	Sodium intake < 2g/d	Sodium intake < 2g/d	24-hurinary sodium	Usual care	38.7	130.3/NA	Non- industry
Huang B[39] (2017)	HD	90	43	55	Cluster randomized controlled trial	6	SBP, DBP	Self-management support, restrict salt intake	Self-management support, restrict salt intake	Dietary salt intake was measured using slightly modified balance formulas: (1) Sodium intake (mol)=[Na+1(mmol/L)×ECW(L)－Na+ 2(mmol/L)×(ECW– IDWG)(L)]/1000; (2) Salt intake (g/day)=sodium intake(mol)×58.4 (g/mol)/2(day)	Usual care	NA	154/86	Non- industry
Ikenoue T[6] (2018)	HD	88115	60	65	Cohort study	12	All-cause mortality, CV events	Salt intake < 6 g/d	Salt intake < 6 g/d	Intra-dialytic weight loss was used to calculate salt intake	Salt intake ≥ 6 g/d	NA	NA	No funding supported
Parvanova A[40] (2018)	Patients with type 2 diabetes and macroalbuminuria	108	89	64	Randomised parallel-group controlled trial	3	UACR	Sodium intake ≤ 100mmol/d	Sodium intake ≤ 100mmol/d	24-hurinary sodium	Sodium intake ≥ 200mmol/d	87.36	146/80	Industry
Martinez MG[41] (2019)	Pre-dialysis CKD with UPE > 500 mg/d	84	52	56	Cohort study	6	Change in SBP	Urinary sodium reduced	Urinary sodium reduced	24-hurinary sodium	Urinary sodium increased	49[b]	142/NA	NA
Nishimoto M[42] (2019)	UACR 30–599 mg/g and eGFR ≥ 50 ml/min/1.73 m^2	304	70	59	Cohort study	12	Change in eGFR, SBP, and DBP	Urinary sodium ≤ 189 mmol/d	Urinary sodium ≤ 189 mmol/d	24-hurinary sodium	Urinary sodium > 189 mmol/d	64.93	140/83	Non- industry

Ahn SY[43] (2019)	eGFR ≥ 30 mL/min/1.73m2 and UACR ≥ 30 mg/g	171	49	50	Cohort study	12	Rate of change in eGFR, renal composite outcomes, change in UAE, SBP, DBP	Intensive low-salt diet education	Intensive low-salt diet education to achieve urinary sodium ≤ 100mmol/d	24-hurinary sodium	Usual care	64.2	124/75	Non- industry
Hu J[44] (2019)	PD	33	58	42	Cohort study	12	Residual renal Ccr	Sodium intake ≤ 3.0 g/d	Sodium intake ≤ 3.0 g/d	Daily sodium intake of the patients was calculated based on the 3-day dietary record	Sodium intake > 3.0 g/d	8.3	146/95	Non- industry
Humalda JK[45] (2020)	CKD stages 1 to 4 or kidney transplants recipients, and eGFR was ≥ 25 mL/min/ 1.73 m2	99	84	57	RCT	6	SBP, DBP	Self-management to restrict salt intake	Self-management to restrict salt intake	A detailed interactive food diary	Usual care	55	139/83	Non- industry
Garofalo C[46] (2020)	CKD stages 1–5	1785	58	64	Cohort study	49	Renal composite outcomes, all-cause mortality	Salt intake < 6 g/d	Salt intake < 6 g/d	24-hurinary sodium	Salt intake ≥ 6 g/d	37	NA	NA
Kinguchi S[47] (2021)	Diabetic kidney disease patients with HbA1c 7.0–10.0%, eGFR ≥ 45 ml/min/1.73 m2, and UACR ≥ 30 mg/g	84	76	65	Cohort study	6	UACR	urinary sodium-to-creatinine ratios < 1.14	Urinary sodium-to-creatinine ratios < 1.14	Urinary sodium-to-creatinine ratios	urinary sodium-to-creatinine ratios ≥ 1.14	67	142/79	Non- industry
Kohatsu K[48] (2021)	CKD stages 3–5	197	76	71	Cohort study	17	Renal composite outcomes, all-cause mortality	Salt intake < 6 g/d	Salt intake < 6 g/d	24-hurinary sodium	Salt intake ≥ 6 g/d	24.2	133/74	Non- industry

Note: Ccr, clearance of creatinine; CKD, chronic kidney disease; CV, cardiovascular; DBP, diastolic blood pressure; eGFR, estimated glomerular filtration rate; ESRD, end stage renal disease; ECW, extracellular water; HD, hemodialysis; IDWG, interdialytic weight gain; NA, no available; PD, peritoneal dialysis; SBP, systolic blood pressure; UACR, urinary albumin to creatinine ratio; UAE, urine albumin excretion; UPE, urinary protein excretion; the units for eGFR: mL/min/1.73m². [a] The units for eGFR: mL/min, [b] Only Ccr obtained, the units for Ccr: mL/min; [c] Only serum creatinine obtained, the units for serum creatinine: μmol/L.